Linear Algebra 2026 Homework 2

Due: October 7, 2026

Problems should be solved in English and calculation steps are required. For all T/F questions, you don't need to justify your answer.

All notations can be found in the book "Linear Algebra — the fourth edition" by Stephen H. Friedberg, Arnold J. Insel, Lawrence E. Spence. If you cannot locate them, feel free to contact the TAs.

Problem 1 [11 pts]. Label the following statements as true or false.

(a) Every vector space contains a zero vector.

(b) A vector space may have more than one zero vector.

(c) In any vector space, $ax = bx$ implies that $a = b$.

(d) In any vector space, $ax = ay$ implies that $x = y$.

(e) A vector in F^n may be regarded as a matrix in $\mathsf{M}_{n\times 1}(\mathsf{F})$.

(f) An $m \times n$ matrix has m columns and n rows.

(g) In $\mathsf{P}(\mathsf{F})$, only polynomials of the same degree may be added.

(h) If f and g are polynomials of degree n, then $f + g$ is a polynomial of degree n.

(i) If f is a polynomial of degree n and c is a nonzero scalar, then cf is a polynomial of degree n.

(j) A nonzero scalar of F may be considered to be a polynomial in $\mathsf{P}(\mathsf{F})$ having degree zero.

(k) Two functions in $\mathcal{F}(S, \mathsf{F})$ are equal if and only if they have the same value at each element of S.

Problem 2 [2 pts]. Write the zero vector of $\mathsf{M}_{3\times 4}(\mathsf{F})$.

Problem 3 [4 pts]. In any vector space V, show that $(a + b)(x + y) = ax + ay + bx + by$ for any $x, y \in \mathsf{V}$ and any $a, b \in \mathsf{F}$.

Problem 4 [4 pts]. Let V denote the set of ordered pairs of real numbers. If (a_1, a_2) and (b_1, b_2) are elements of V and $c \in R$, define

$$(a_1, a_2) + (b_1, b_2) = (a_1 + b_1, a_2 b_2) \quad \text{and} \quad c(a_1, a_2) = (ca_1, a_2).$$

Is V a vector space over R with these operations? Justify your answer.

Problem 5 [6 pts]. Let V and W be vector spaces over a field F. Let

$$\mathsf{Z} = \{(v, w) : v \in \mathsf{V} \text{ and } w \in \mathsf{W}\}.$$

Prove that Z is a vector space over F with the operations

$$(v_1, w_1) + (v_2, w_2) = (v_1 + v_2, w_1 + w_2) \quad \text{and} \quad c(v_1, w_1) = (cv_1, cw_1).$$

Problem 6 [7 pts]. Label the following statements as true or false.

(a) If V is a vector space and W is a subset of V that is a vector space, then W is a subspace of V.

(b) The empty set is a subspace of every vector space.

(c) If V is a vector space other than the zero vector space, then V contains a subspace W such that $\mathsf{W} \neq \mathsf{V}$.

(d) The intersection of any two subsets of V is a subspace of V.

(e) An $n \times n$ diagonal matrix can never have more than n nonzero entries.

(f) The trace of a square matrix is the product of its diagonal entries.

(g) Let W be the xy-plane in R^3; that is, $\mathsf{W} = \{(a_1, a_2, 0) : a_1, a_2 \in R\}$. Then $\mathsf{W} = \mathsf{R}^2$.

Problem 7 [6 pts]. Determine whether the following sets are subspaces of R^3 under the operations of addition and scalar multiplication defined on R^3. Justify your answers.

(a) $\mathsf{W}_1 = \{(a_1, a_2, a_3) \in \mathsf{R}^3 : a_1 = 3a_2 \text{ and } a_3 = -a_2\}$

(b) $\mathsf{W}_2 = \{(a_1, a_2, a_3) \in \mathsf{R}^3 : a_1 = a_3 + 2\}$

Problem 8 [6 pts]. Prove that $\mathsf{W}_1 = \{(a_1, a_2, \ldots, a_n) \in \mathsf{F}^n : a_1 + a_2 + \cdots + a_n = 0\}$ is a subspace of F^n, but $\mathsf{W}_2 = \{(a_1, a_2, \ldots, a_n) \in \mathsf{F}^n : a_1 + a_2 + \cdots + a_n = 1\}$ is not.

Problem 9 [10 pts]. Let W_1 and W_2 be subspaces of a vector space V. Prove that $\mathsf{W}_1 \cup \mathsf{W}_2$ is a subspace of V if and only if $\mathsf{W}_1 \subseteq \mathsf{W}_2$ or $\mathsf{W}_2 \subseteq \mathsf{W}_1$.

The following definitions are needed for Problem 10.

Definition 1. If S_1 and S_2 are nonempty subsets of a vector space V, then the **sum** of S_1 and S_2, denoted $S_1 + S_2$, is the set $\{x + y : x \in S_1 \text{ and } y \in S_2\}$.

Definition 2. A vector space V is called the **direct sum** of W_1 and W_2 if W_1 and W_2 are subspaces of V such that $\mathsf{W}_1 \cap \mathsf{W}_2 = \{0\}$ and $\mathsf{W}_1 + \mathsf{W}_2 = \mathsf{V}$. We denote that V is the direct sum of W_1 and W_2 by writing $\mathsf{V} = \mathsf{W}_1 \oplus \mathsf{W}_2$.

Problem 10 [10 pts]. Let W_1 and W_2 be subspaces of a vector space V.

(a) Prove that $\mathsf{W}_1 + \mathsf{W}_2$ is a subspace of V that contains both W_1 and W_2.

(b) Prove that any subspace of V that contains both W_1 and W_2 must also contain $\mathsf{W}_1 + \mathsf{W}_2$.

Problem 11 [20 pts]. Let W be a subspace of a vector space V over a field F. For any $v \in \mathsf{V}$ the set $\{v\} + \mathsf{W} = \{v + w : w \in \mathsf{W}\}$ is called the **coset** of W containing v. It is customary to denote this coset by $v + \mathsf{W}$ rather than $\{v\} + \mathsf{W}$.

(a) Prove that $v + \mathsf{W}$ is a subspace of V if and only if $v \in \mathsf{W}$.

(b) Prove that $v_1 + \mathsf{W} = v_2 + \mathsf{W}$ if and only if $v_1 - v_2 \in \mathsf{W}$.

Addition and scalar multiplication by scalars of F can be defined in the collection $S = \{v + \mathsf{W} : v \in \mathsf{V}\}$ of all cosets of W as follows:

$$(v_1 + \mathsf{W}) + (v_2 + \mathsf{W}) = (v_1 + v_2) + \mathsf{W}$$

for all $v_1, v_2 \in \mathsf{V}$ and

$$a(v + \mathsf{W}) = av + \mathsf{W}$$

for all $v \in \mathsf{V}$ and $a \in F$.

(c) Prove that the preceding operations are well defined; that is, show that if $v_1 + \mathsf{W} = v_1' + \mathsf{W}$ and $v_2 + \mathsf{W} = v_2' + \mathsf{W}$, then

$$(v_1 + \mathsf{W}) + (v_2 + \mathsf{W}) = (v_1' + \mathsf{W}) + (v_2' + \mathsf{W})$$

and

$$a(v_1 + \mathsf{W}) = a(v_1' + \mathsf{W})$$

for all $a \in F$.

(d) Prove that the set S is a vector space with the operations defined in (c). This vector space is called the **quotient space of V modulo W** and is denoted by V/W.

Problem 12 [6 pts]. Label the following statements as true or false.

(a) If S is a linearly dependent set, then each vector in S is a linear combination of other vectors in S.

(b) Any set containing the zero vector is linearly dependent.

(c) The empty set is linearly dependent.

(d) Subsets of linearly dependent sets are linearly dependent.

(e) Subsets of linearly independent sets are linearly independent.

(f) If $a_1x_1 + a_2x_2 + \cdots + a_nx_n = 0$ and $x_1, x_2, \ldots, x_n$ are linearly independent, then all the scalars a_i are zero.

Problem 13 [6 pts]. Determine whether the following sets are linearly dependent or linearly independent.

(a) $\left\{ \begin{pmatrix} 1 & -3 \\ -2 & 4 \end{pmatrix}, \begin{pmatrix} -2 & 6 \\ 4 & -8 \end{pmatrix} \right\}$ in $\mathsf{M}_{2\times 2}(R)$

(c) $\{x^3 + 2x^2, -x^2 + 3x + 1, x^3 - x^2 + 2x - 1\}$ in $\mathsf{P}_3(R)$

Problem 14 [6 pts]. In $\mathsf{M}_{m\times n}(F)$, let E^{ij} denote the matrix whose only nonzero entry is 1 in the ith row and jth column. Prove that $\{E^{ij} : 1 \le i \le m,\ 1 \le j \le n\}$ is linearly independent.

Problem 15 [10 pts]. Let $S = \{u_1, u_2, \ldots, u_n\}$ be a linearly independent subset of a vector space V over the field Z_2. How many vectors are there in span(S)? Justify your answer.

Problem 16 [10 pts]. Prove that if $\{A_1, A_2, \ldots, A_k\}$ is a linearly independent subset of $\mathsf{M}_{n\times n}(F)$, then $\{A_1^t, A_2^t, \ldots, A_k^t\}$ is also linearly independent.

Problem 17 [10 pts]. Let $f, g \in \mathcal{F}(R, R)$ be the functions defined by $f(t) = e^{rt}$ and $g(t) = e^{st}$, where $r \neq s$. Prove that f and g are linearly independent in $\mathcal{F}(R, R)$.

Problem 18 [6 pts]. Determine which of the following sets are bases for R^3.

(a) $\{(1, 0, -1), (2, 5, 1), (0, -4, 3)\}$

(b) $\{(2, -4, 1), (0, 3, -1), (6, 0, -1)\}$

Problem 19 [4 pts]. Is $\{(1, 4, -6), (1, 5, 8), (2, 1, 1), (0, 1, 0)\}$ a linearly independent subset of R^3? Justify your answer.

Problem 20 [16 pts]. Consider a system of linear equations

$$\begin{aligned}
(a_1 + b)x_1 + a_2x_2 + a_3x_3 + \cdots + a_nx_n &= 0\\
a_1x_1 + (a_2 + b)x_2 + a_3x_3 + \cdots + a_nx_n &= 0\\
a_1x_1 + a_2x_2 + (a_3 + b)x_3 + \cdots + a_nx_n &= 0\\
&\cdots\\
a_1x_1 + a_2x_2 + a_3x_3 + \cdots + (a_n + b)x_n &= 0
\end{aligned}$$

where $\sum_{i=1}^{n} a_i \neq 0$.

Determine the relationship between $a_1, a_2, \cdots, a_n$ and b such that

(a) This linear system has only a zero solution. (8 pts)

(b) This system has nonzero solutions. In this case, determine a solution set. (8 pts)